

## **Rouge Construction Fault Tolerance Implementations**

College of Science, Engineering, and Technology, Grand Canyon University

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Instructor Joel

Liéla Pressley

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## **Rouge Construction Fault Tolerance Implementations**

Fault Tolerance is implementing hardware connections and setups that will allow a network to maintain connection despite failure or faults somewhere on the network. Rouge Construction has many moving parts and to provide some fault tolerance while remaining cost-efficient, the key elements applied include link aggregation between switch and routers, T1 Redundant Connectivity, and Common Address Redundancy Protocol on a partial mesh subnet.

An element implemented that allows for automatic failover between NICs, load balancing, and increased throughput between switches and routers is link aggregation. Since each network within Rouge Construction has important data that needs consistent transmission with high traffic levels this is a great strategy. The Wide Area Network for the warehouse, the Operation and Supply Chain Management office subnet, Virtual Area Network for the HR department and the IPv6 network for the three construction/concrete offices have link aggregations.

Since the Administration office upholds important data and traffic, it would be devastating for the company as a whole if it experiences failure. To help prevent a single point of failure, T1 Redundant Connectivity was applied to its network. An additional firewall, router, and switch were installed to provide a second pathway that data can be transmitted to clients adding a layer of redundancy. The customer services and data office has a partial mesh topology, to accommodate its structure, CARP (Common Address Redundancy Protocol) was implemented. This protocol allows for a group of computers to share one IP address and be a group of redundancy. There is one computer acting as group master which receives requests on behalf of the others and parcels as needed. Since these clients receive, transmit, and handle sensitive customer data it's important to have it be secure and redundant.